

# DHCP REDESIGN

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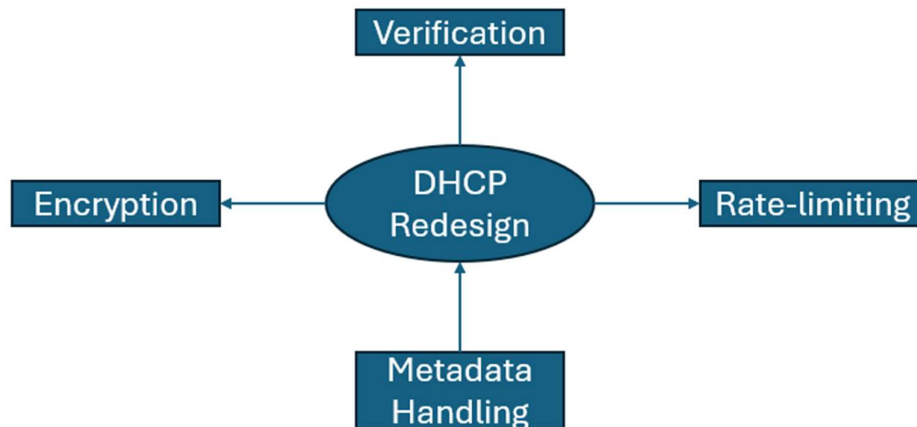
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## Introduction

This DHCP redesign was designed make DHCP more efficient and reliable compared to current methods. This DHCP redesign was made because DHCP is one of the most local network protocols. This is because it allocates IP addresses which are used for routing and communication on internet and within local networks. This redesign was made to ensure that is it is secure so it can function reliably even if attacked. This DHCP redesign does not replace or remove the message system or how IP addresses are managed in memory. Instead, this DHCP redesign focuses on encryption, authentication, metadata and DHCP discovery.

This DHCP redesign cannot promise perfect security and efficiency as no system ever can. However, this DHCP redesign can promise to try improving DHCP as much as possible. This DHCP redesign was targeted towards modern home networks with some features being able to support legacy systems. While the technical specifications in this document are for network engineers. Make sure to adapt this system as needed to real world standards.



The spider diagram above shows the 4 main concepts involved in this DHCP redesign. These 4 main concepts are mainly security changes as DHCP in nature is minimal and efficient. You can find more information on these concepts in the redesign below.

## **DHCP Redesign Main Section**

- The first change is adding built-in encryption.
- This can be achieved by using pre-shared keys or a pre-query key to encrypt each packet individually.
- Each packet must be encrypted individually as UDP is connectionless meaning no persistent session to establish a handshake.
- Pre-shared keys are recommended for home networks as they are suitable because of their simplicity and basic setup.
- Pre-query keys are recommended for enterprise or high-risk networks as they are more secure and constantly rotating.
- Built-in encryption is important as it ensures data is protected during transit.
- The second change is a verification system.
- First, each device on the network is given a public key to use to verify the signature of the DHCP server.
- Second, the DHCP server is signed using a private key which is user-configured.
- Third, devices on the network use their public key when a server responds to their discovery message to verify the authenticity of the server.
- Fourth, if the signature breaks, then the device will send out a new broadcast repeating this process until it finds the correct server.
- The private key should be stored securely on the DHCP server and can be changed if compromise has been detected.
- This verification system is important because it ensures that devices use the correct DHCP server instead of a malicious server.
- The third change is metadata handling
- This can be achieved by including metadata into the encrypted payload or reducing the amount of metadata collected.
- For limiting metadata, the following list should be removed for that option Hostname, Vendor class identifier, parameter request list and client identifier.
- For encrypting metadata, only least time, hostname, vendor class, parameter request can be encrypted.
- Essential metadata such as IP address, subnet mask, default gateway, DNS servers, MAC address and server identifier is needed for routing and networking meaning it cannot be encrypted.
- The fourth change is rate-limiting
- This can be achieved by allowing users to configure how many requests an DHCP server can receive from a device in a duration. This is an optional setting
- The fifth change is DHCP discover format.
- This can be achieved by the client including what type of IP address they need from the DHCP server.
- If the server supports this updated format, it will use it, while legacy systems can be configured to ignore it.

## **DHCP Redesign Trade-offs**

- This DHCP Redesign has trade-offs just like any other system I have created.
- There may also be further implementation challenges that engineers and adopters may face.
- I have identified 5 trade-offs.
- The first trade-off is performance.
- This trade-off exists due to the overhead that encryption and verification have on the speed of DHCP.
- This is because each client must ensure that data is encrypted and verify the DHCP server before they can get an IP address.
- This trade-off can be managed by using quick but secure encryption algorithms and accepting that DHCP won't be as fast as it used to be.
- The second trade-off is time-consumption.
- This trade-off exists as developing an DHCP protocol with encryption, verification and multiple options is time-consuming.
- This means that development could take years as this is a full redesign and not a patch.
- This trade-off can be managed by using structured project management processes, pre-existing systems and avoiding unnecessary complexity.
- The third trade-off is cost.
- This trade-off exists as it will be expensive to hire the engineers needed to do this and test the protocol across systems.
- This is because this is a project that will take years, require legacy systems to be tested and to ensure it doesn't break the internet.
- This trade-off can be managed by allocating a budget, using simulations for testing and accepting that every system comes with a cost.
- The fourth trade-off is compatibility
- This trade-off exists because not every system will be able to support this redesign.
- This is because legacy systems do not have the capability needed to support the security features of this redesign.
- This trade-off can be managed by building in fallbacks for legacy systems, not forcing legacy systems to use this protocol if it breaks and accepting compatibility won't be perfect.
- The fifth trade-off is key management.
- This trade-off exists because since DHCP uses multiple encryption keys they must be managed securely.
- This is because if the keys are exposed or compromised the exchanges are vulnerable.
- This trade-off can be managed by storing the keys in secure methods like TPM, HSM, encrypted files or a password manager.